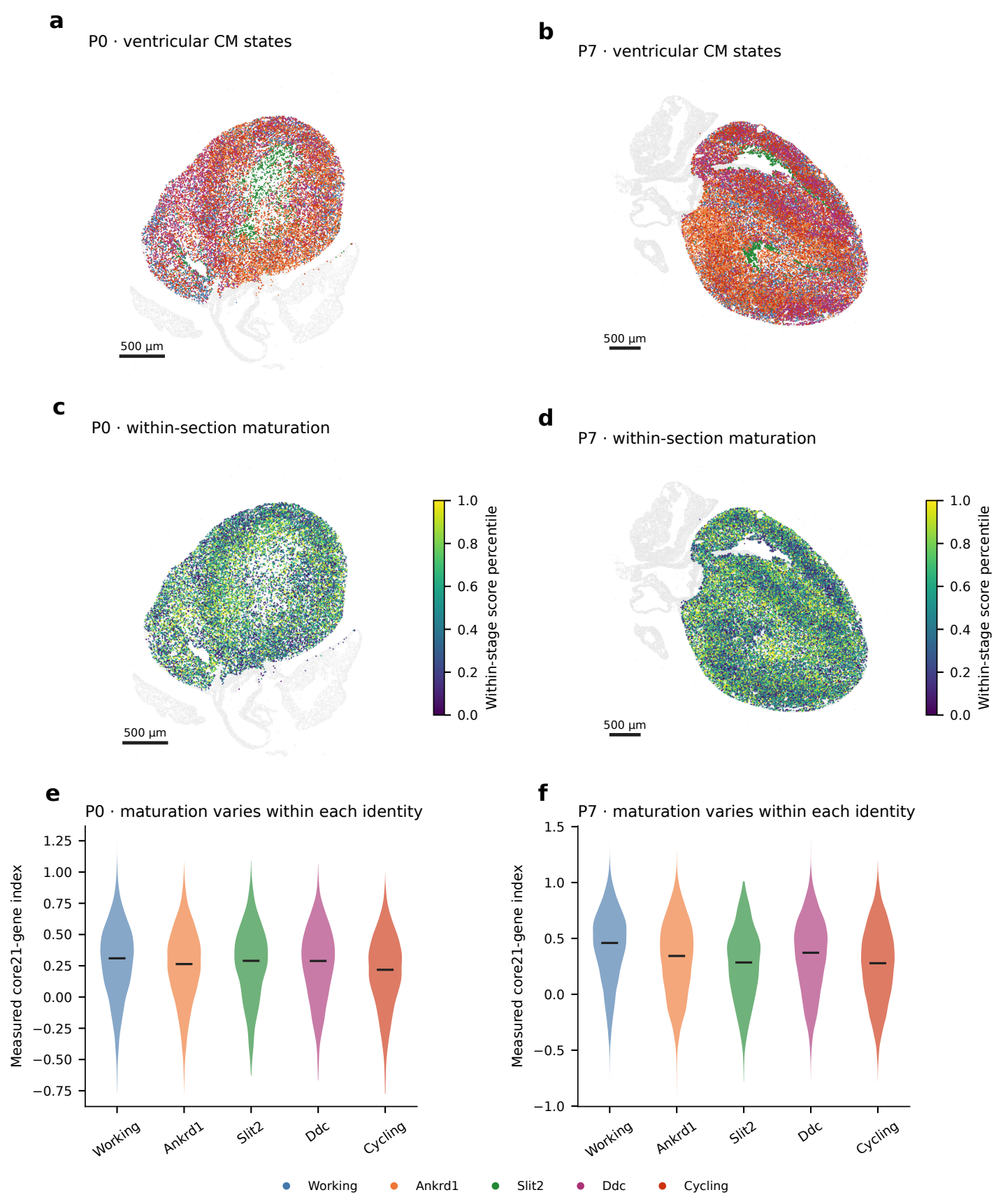
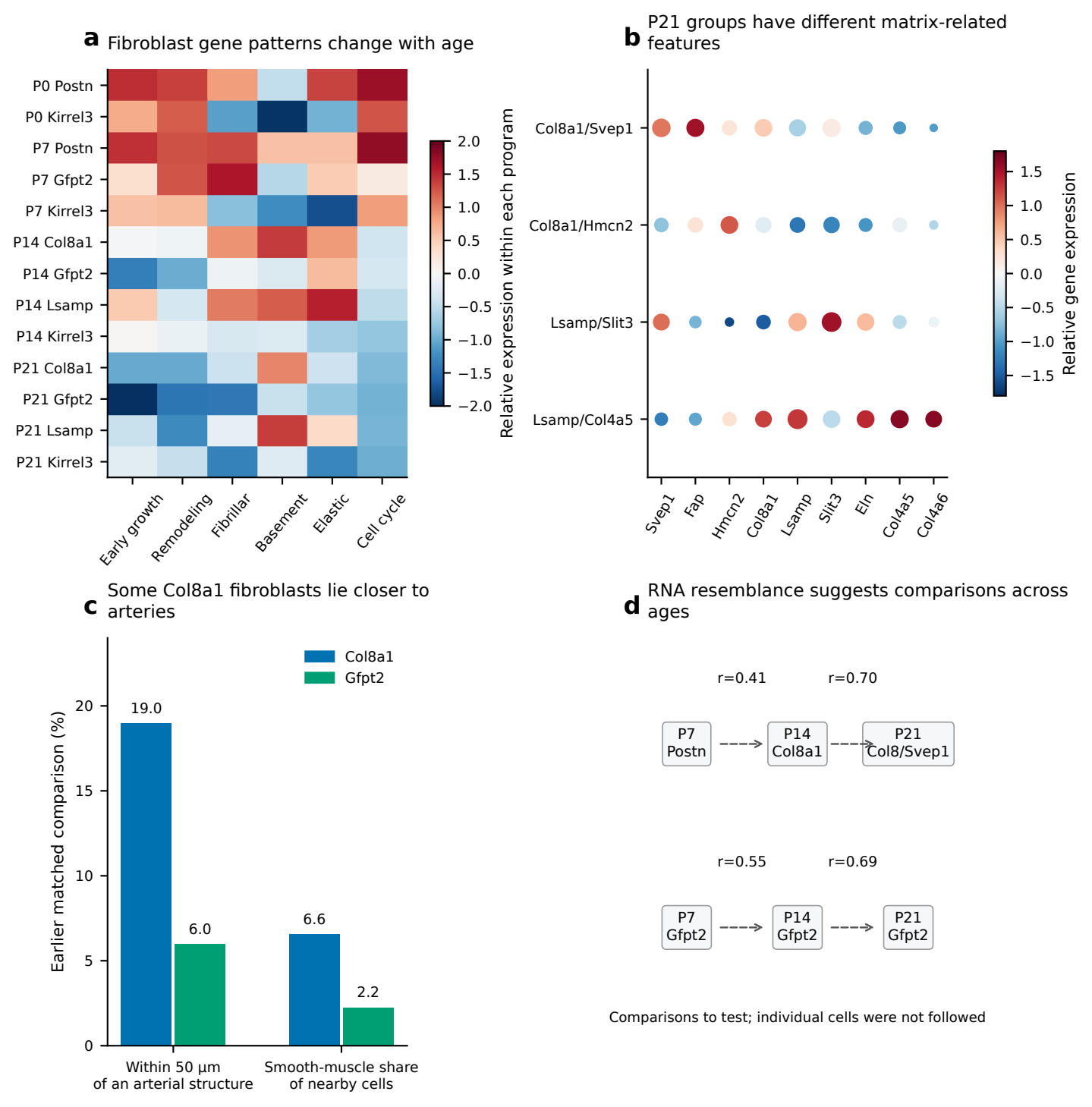
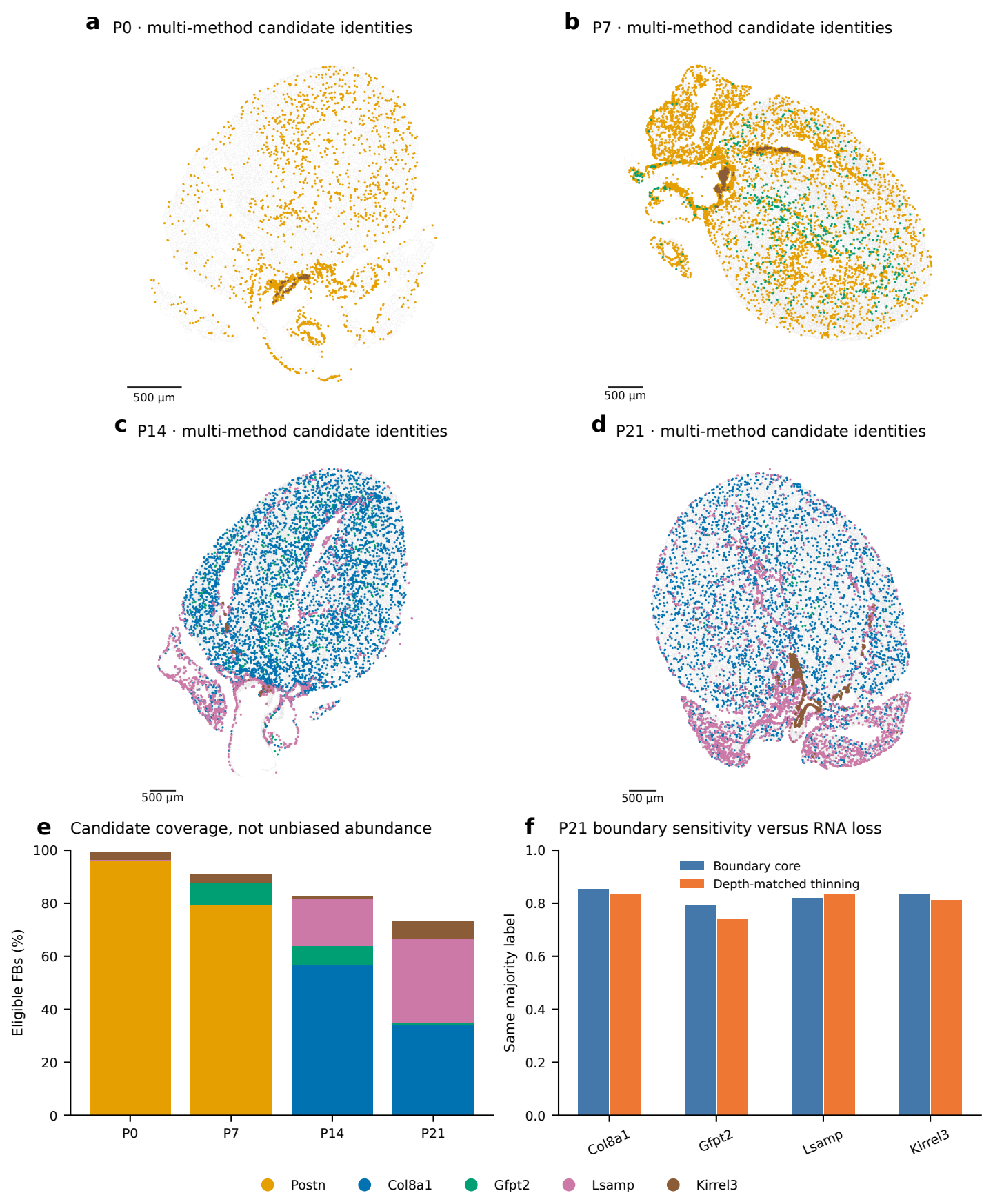
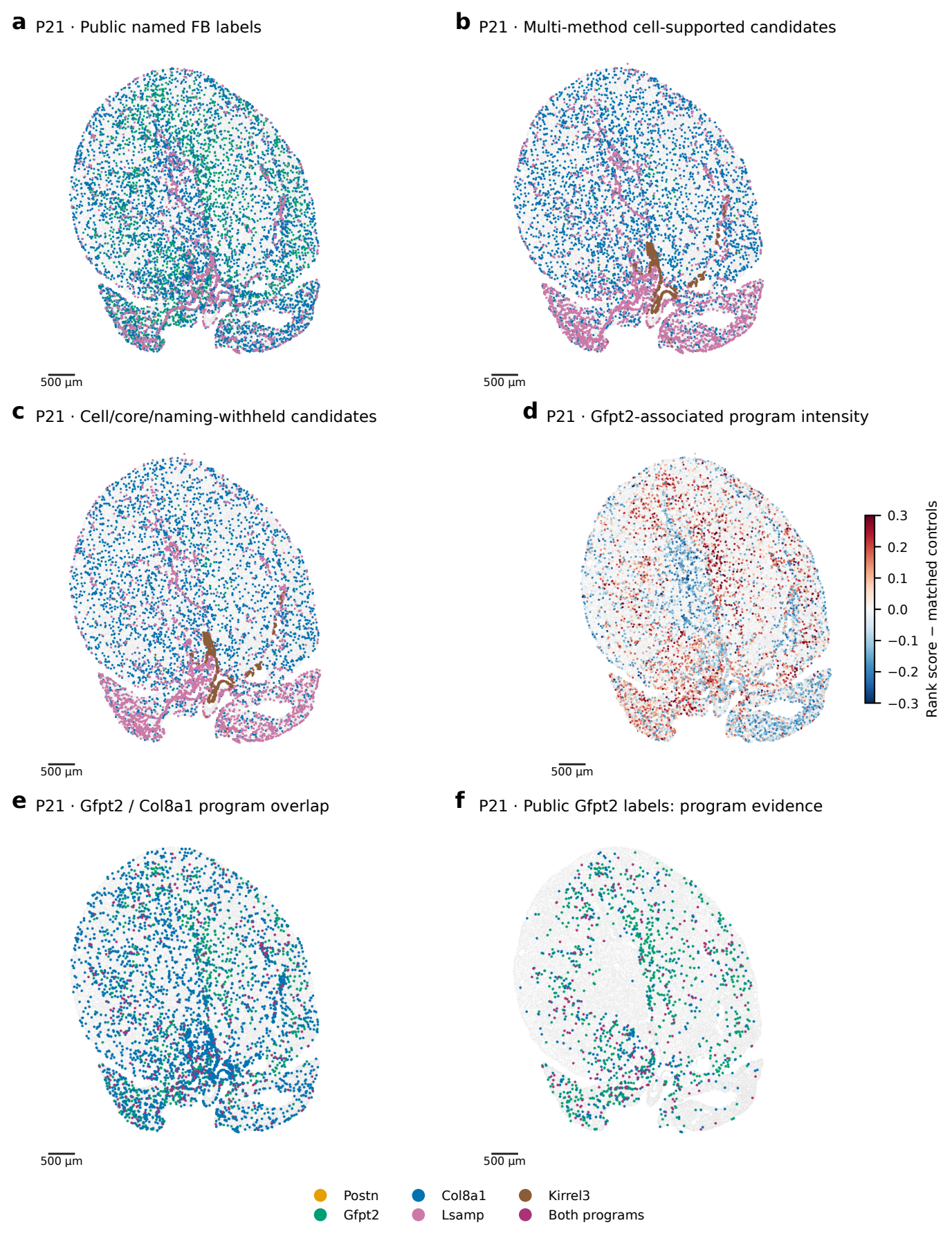
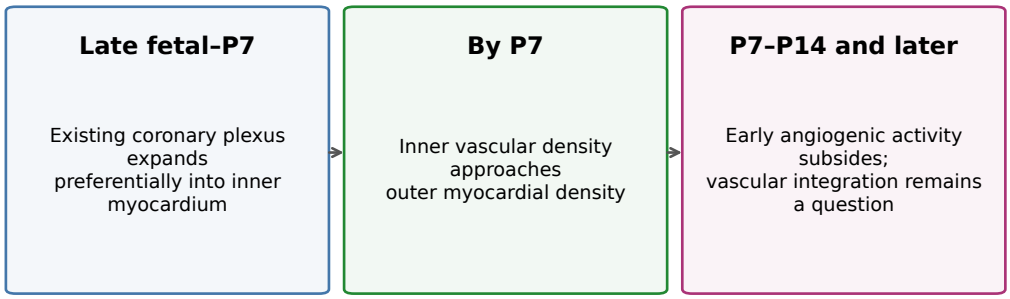

a
P0 · ventricular CM states
500 µm
b
P7 · ventricular CM states
500 µm
c
P0 · within-section maturation
500 µm
1.0
0.8
0.6
0.4
0.2
0.0
Within-stage score percentile
d
P7 · within-section maturation
500 µm
1.0
0.8
0.6
0.4
0.2
0.0
Within-stage score percentile
e
P0 · maturation varies within each identity
Measured core21-gene index
1.25
1.00
0.75
0.50
0.25
0.00
−0.25
−0.50
−0.75
Working
Ankrd1
Slit2
Ddc
Cycling
f
P7 · maturation varies within each identity
Measured core21-gene index
1.5
1.0
0.5
0.0
−0.5
−1.0
Working
Ankrd1
Slit2
Ddc
Cycling
Working
Ankrd1
Slit2
Ddc
Cycling


a
Fibroblast gene patterns change with age
P0 Postn
P0 Kirrel3
P7 Postn
P7 Gfpt2
P7 Kirrel3
P14 Col8a1
P14 Gfpt2
P14 Lsamp
P14 Kirrel3
P21 Col8a1
P21 Gfpt2
P21 Lsamp
P21 Kirrel3
Early growth
Remodeling
Fibrillar
Basement
Elastic
Cell cycle
2.0
1.5
1.0
0.5
0.0
−0.5
−1.0
−1.5
−2.0
b
P21 groups have different matrix-related features
Relative expression within each program
Col8a1/Svep1
Col8a1/Hmcn2
Lsamp/Slit3
Lsamp/Col4a5
Svep1
Fap
Hmcn2
Col8a1
Lsamp
Slit3
Eln
Col4a5
Col4a6
1.5
1.0
0.5
0.0
−0.5
−1.0
−1.5
Relative gene expression
c
Some Col8a1 fibroblasts lie closer to arteries
Col8a1
Gfpt2
Earlier matched comparison (%)
20
15
10
5
0
19.0
6.0
6.6
2.2
Within 50 µm of an arterial structure
Smooth-muscle share of nearby cells
d
RNA resemblance suggests comparisons across ages
r=0.41
r=0.70
P7 Postn
P14 Col8a1
P21 Col8/Svep1
r=0.55
r=0.69
P7 Gfpt2
P14 Gfpt2
P21 Gfpt2
Comparisons to test; individual cells were not followed


a
P0 · multi-method candidate identities
500 µm
b
P7 · multi-method candidate identities
500 µm
c
P14 · multi-method candidate identities
500 µm
d
P21 · multi-method candidate identities
500 µm
e
Candidate coverage, not unbiased abundance
Eligible FBs (%)
100
80
60
40
20
0
P0
P7
P14
P21
f
P21 boundary sensitivity versus RNA loss
Same majority label
1.0
0.8
0.6
0.4
0.2
0.0
Boundary core
Depth-matched thinning
Col8a1
Gfpt2
Lsamp
Kirrel3
Postn
Col8a1
Gfpt2
Lsamp
Kirrel3

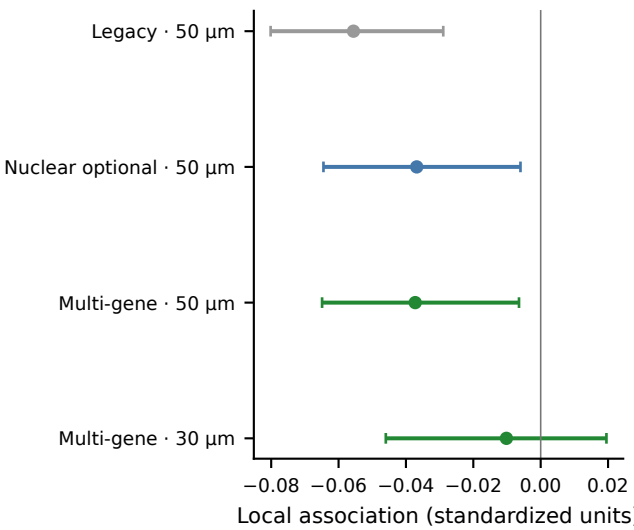

a
P21 · Public named FB labels
500 µm
b
P21 · Multi-method cell-supported candidates
500 µm
c
P21 · Cell/core/naming-withheld candidates
500 µm
d
P21 · Gfpt2-associated program intensity
500 µm
0.3
0.2
0.1
0.0
−0.1
−0.2
−0.3
Rank score − matched controls
e
P21 · Gfpt2 / Col8a1 program overlap
500 µm
f
P21 · Public Gfpt2 labels: program evidence
500 µm
Postn
Gfpt2
Col8a1
Lsamp
Kirrel3
Both programs

a Regional vascular expansion changes the setting in which CMs develop

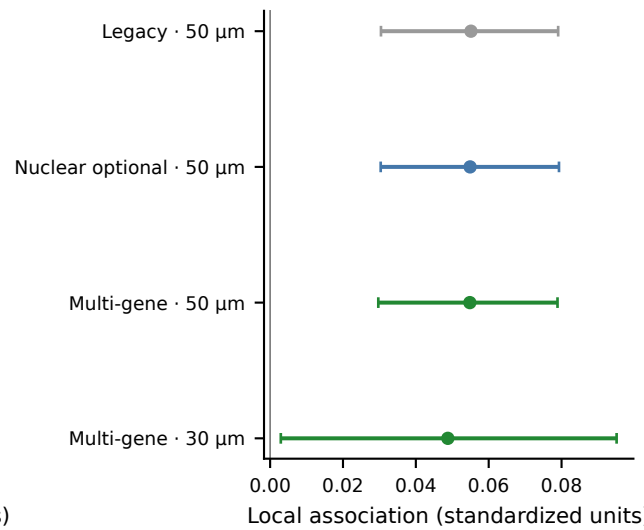


Published vascular findings (Lu et al., 2021); not newly measured perfusion in our sections

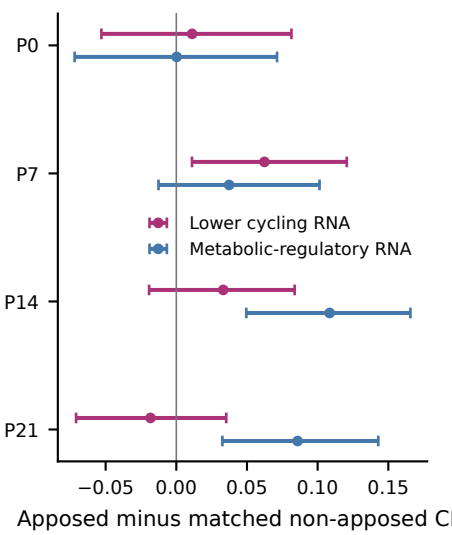
b P7 cycling CMs · Postn FB / metabolic RNA



c P14 Ankrd1 CMs · endocardial EC / sarcomere RNA



d Capillary apposition · different components over time



e Growth, access and functional integration are distinct

Growing vasculature

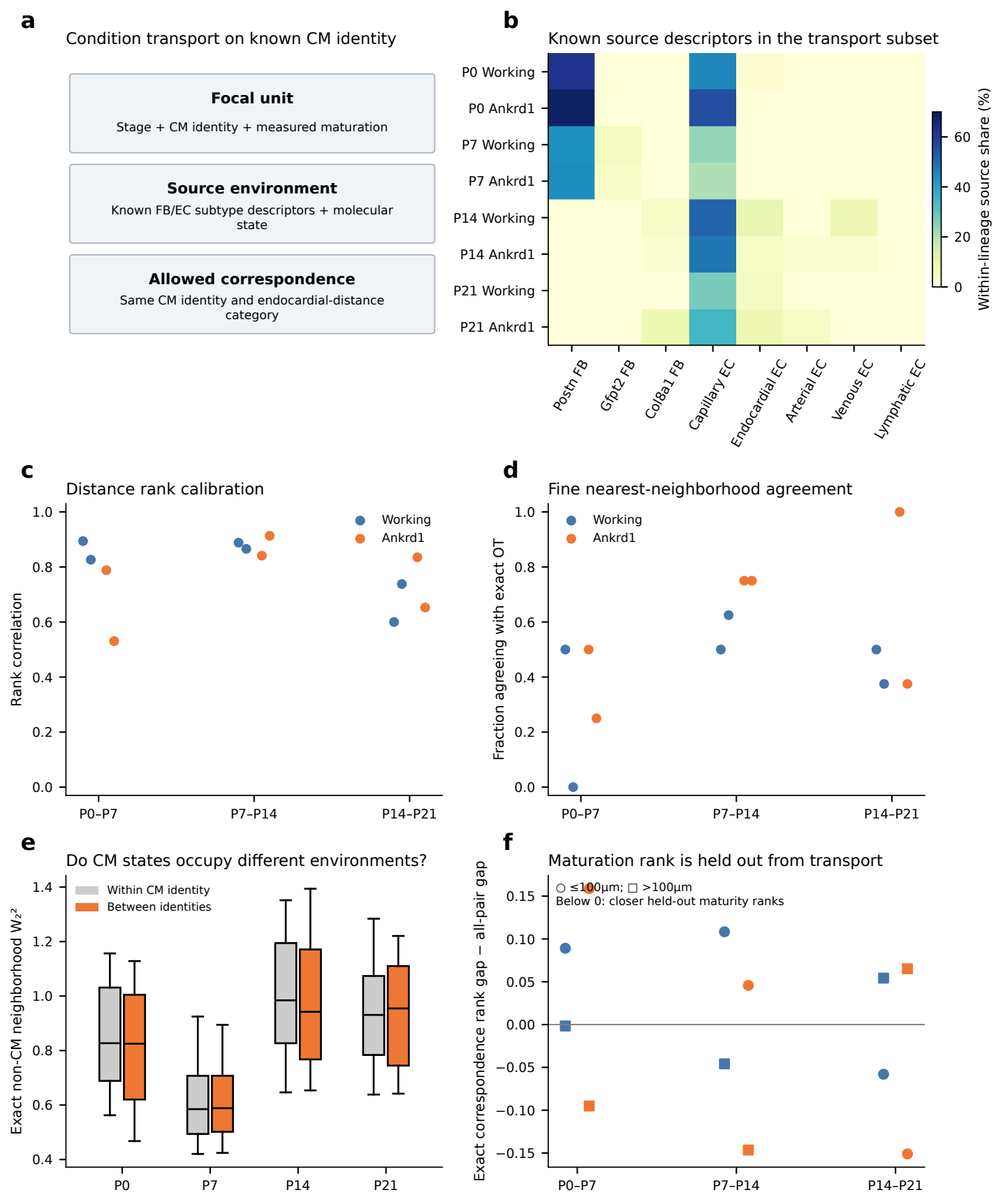
EC cycling / sprouting-associated RNA; not measured oxygen delivery

Local vascular access

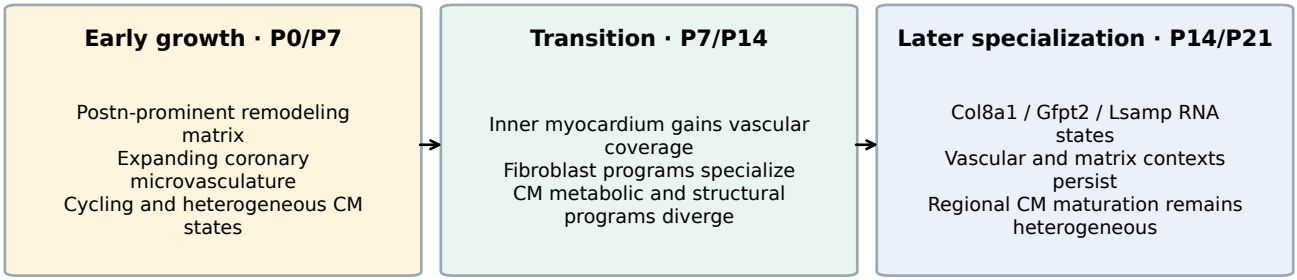
Capillary distance or boundary apposition; not proof of a perfused vessel

CM developmental response

Metabolism, sarcomeres and cycling can differ; functional maturation requires direct measurements

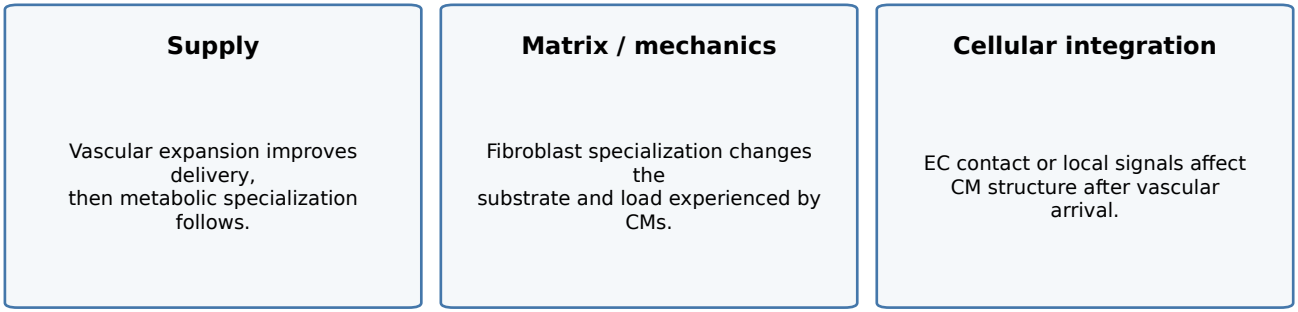

a
Condition transport on known CM identity
Focal unit
Stage + CM identity + measured maturation
Source environment
Known FB/EC subtype descriptors + molecular state
Allowed correspondence
Same CM identity and endocardial-distance category
b
Known source descriptors in the transport subset
P0 Working
P0 Ankrd1
P7 Working
P7 Ankrd1
P14 Working
P14 Ankrd1
P21 Working
P21 Ankrd1
Postn FB
Gfpt2 FB
Col8a1 FB
Capillary EC
Endocardial EC
Arterial EC
Venous EC
Lymphatic EC
Within-lineage source share (%)
0
20
40
60
c
Distance rank calibration
Working
Ankrd1
Rank correlation
P0–P7
P7–P14
P14–P21
d
Fine nearest-neighborhood agreement
Fraction agreeing with exact OT
e
Do CM states occupy different environments?
Within CM identity
Between identities
Exact non-CM neighborhood W₂²
P0
P7
P14
P21
f
Maturation rank is held out from transport
○ ≤100µm; □ >100µm
Below 0: closer held-out maturity ranks
Exact correspondence rank gap − all-pair gap

a A developmental model: coordinated changes with different local timing



Proposed synthesis; arrows denote developmental windows, not observed cell conversion or a causal chain

b Three competing explanations for regional CM development



Ankrd1 and working CMs may respond differently within the same anatomical region

c Experiments that distinguish these explanations

P3/P5/P10 in independent hearts

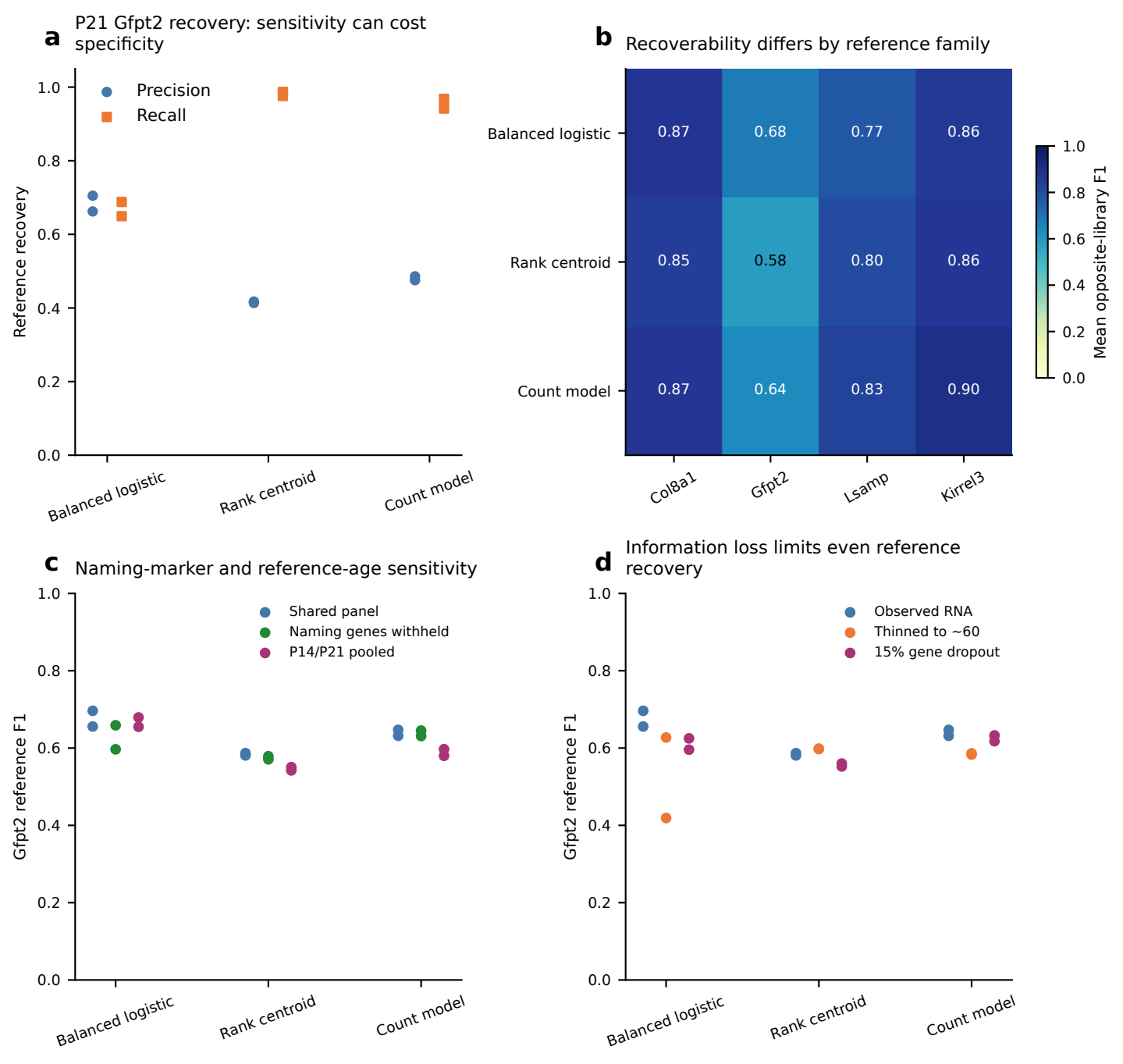
Co-measure perfusion, hypoxia, CM state and matrix organization; resolve temporal order.

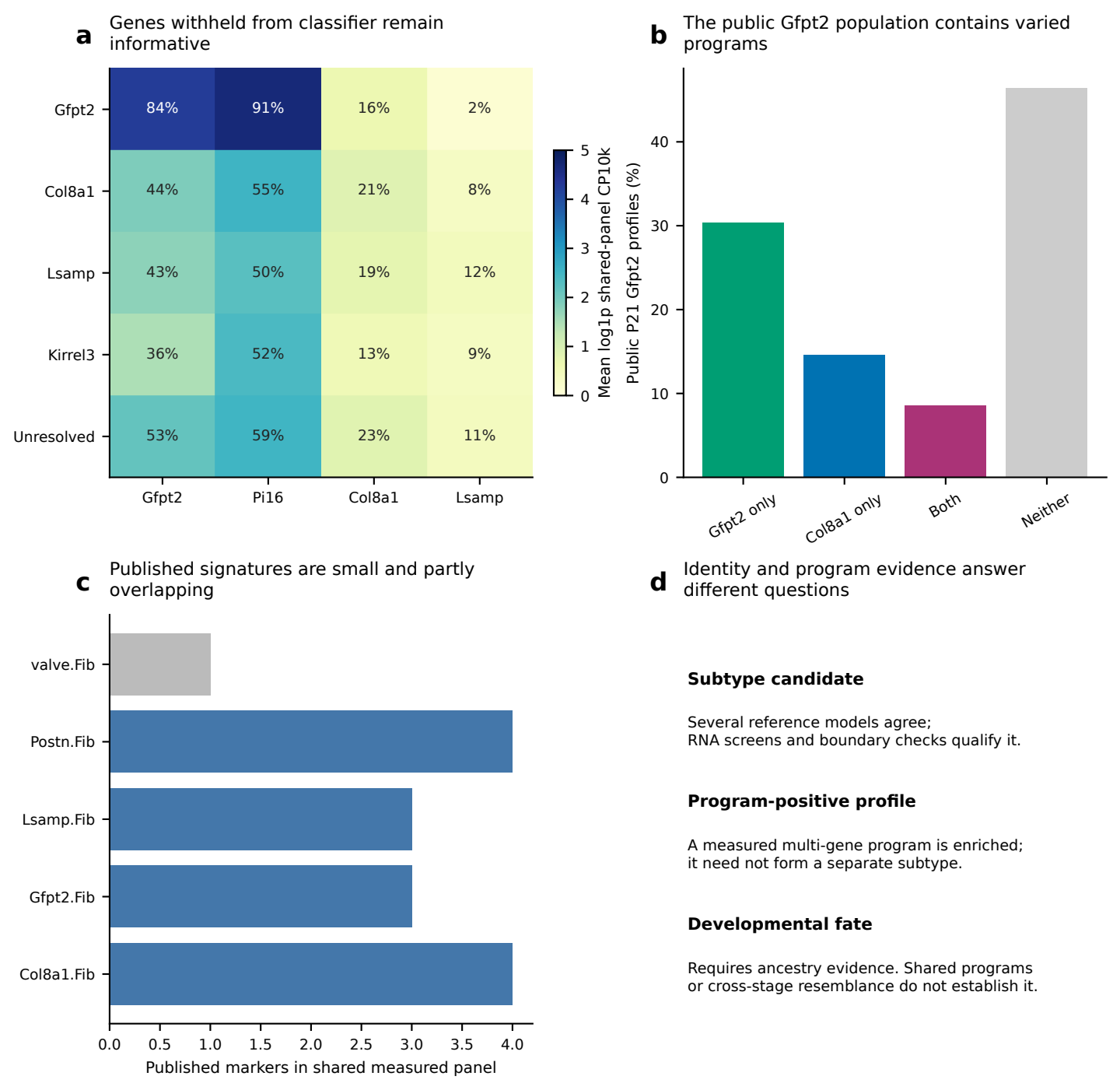
Match the starting CM identity

Separate perfusion/oxygen, EC contact and fibroblast-derived matrix; measure respiration, calcium and force.

Trace proposed state continuity

Test early-to-late fibroblast ancestry; use transport to nominate comparisons, not to establish fate.


a
P21 Gfpt2 recovery: sensitivity can cost specificity
Precision
Recall
Reference recovery
1.0
0.8
0.6
0.4
0.2
0.0
Balanced logistic
Rank centroid
Count model
b
Recoverability differs by reference family
Balanced logistic
Rank centroid
Count model
0.87
0.68
0.77
0.86
0.85
0.58
0.80
0.86
0.87
0.64
0.83
0.90
Col8a1
Gfpt2
Lsamp
Kirrel3
Mean opposite-library F1
1.0
0.8
0.6
0.4
0.2
0.0
c
Naming-marker and reference-age sensitivity
Shared panel
Naming genes withheld
P14/P21 pooled
Gfpt2 reference F1
1.0
0.8
0.6
0.4
0.2
0.0
Balanced logistic
Rank centroid
Count model
d
Information loss limits even reference recovery
Observed RNA
Thinned to ~60
15% gene dropout
Gfpt2 reference F1
1.0
0.8
0.6
0.4
0.2
0.0
Balanced logistic
Rank centroid
Count model


a
Genes withheld from classifier remain informative
Gfpt2
Col8a1
Lsamp
Kirrel3
Unresolved
84%
91%
16%
2%
44%
55%
21%
8%
43%
50%
19%
12%
36%
52%
13%
9%
53%
59%
23%
11%
Gfpt2
Pi16
Col8a1
Lsamp
Mean log1p shared-panel CP10k
0
1
2
3
4
5
b
The public Gfpt2 population contains varied programs
Public P21 Gfpt2 profiles (%)
0
10
20
30
40
Gfpt2 only
Col8a1 only
Both
Neither
c
Published signatures are small and partly overlapping
valve.Fib
Postn.Fib
Lsamp.Fib
Gfpt2.Fib
Col8a1.Fib
0.0
0.5
1.0
1.5
2.0
2.5
3.0
3.5
4.0
Published markers in shared measured panel
d
Identity and program evidence answer different questions
Subtype candidate
Several reference models agree;
RNA screens and boundary checks qualify it.
Program-positive profile
A measured multi-gene program is enriched;
it need not form a separate subtype.
Developmental fate
Requires ancestry evidence. Shared programs
or cross-stage resemblance do not establish it.